Behavioral evolution

IB 431 – Behavioral Ecology

Fall 2025

September 2nd 2025

Announcements

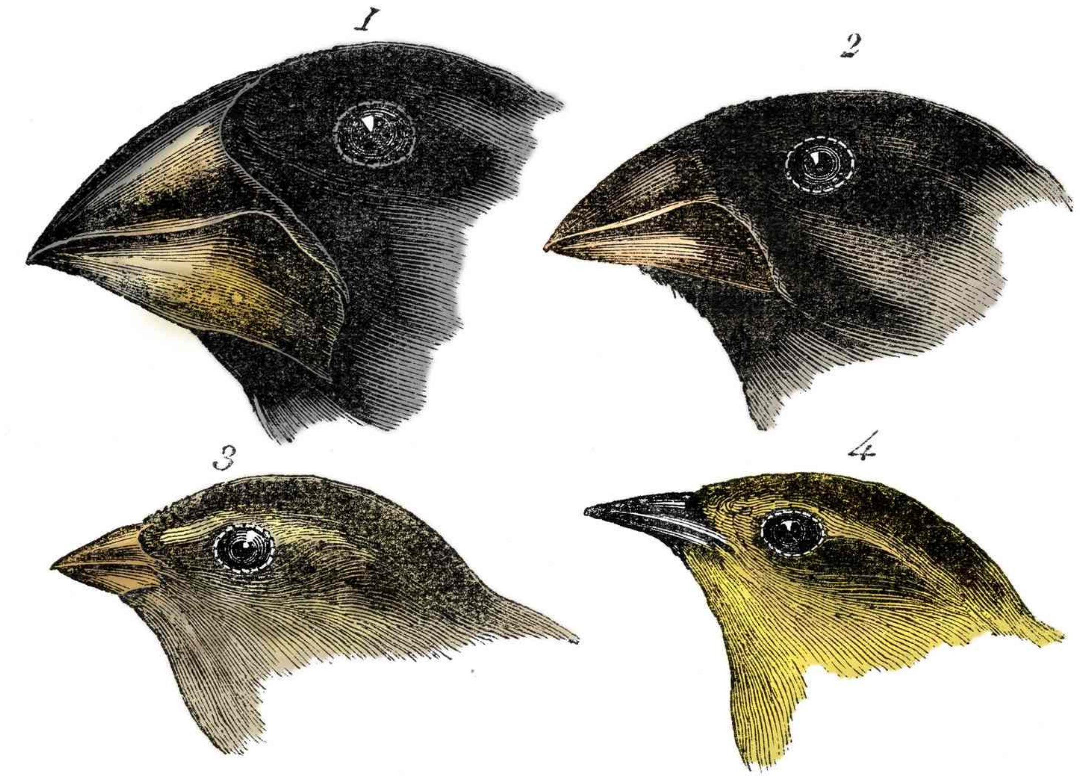
- Sign up to lead discussions if you have not done so
- Attending seminars for extra credit – see class website for list of potential seminars

Outline

I. Review of evolution and natural selection
II. Why study behavioral evolution?
III. How can we study behavioral evolution?
IV. Case studies
V. Go over instructions for discussion leaders
VI. Discuss Trut 1999

Evolution

- What is evolution?
- What is natural selection?
- What is the difference between them?

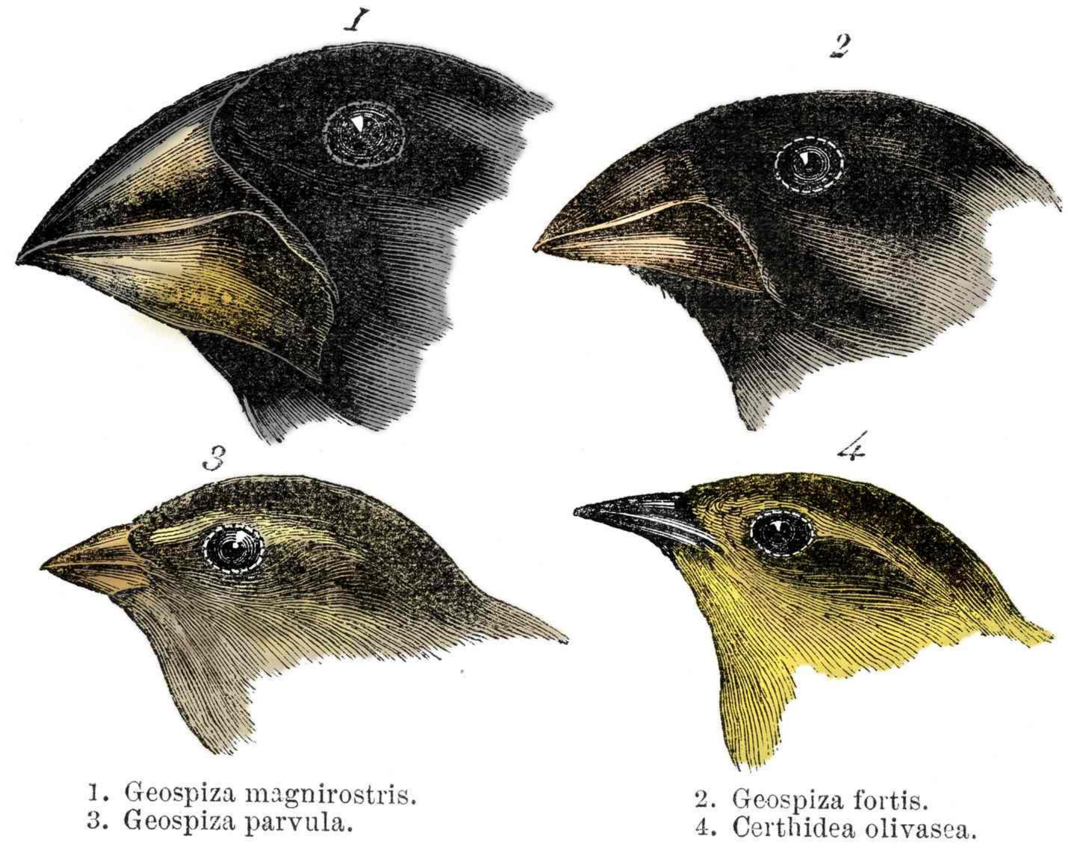


1. Geospiza magnirostris.
3. Geospiza parvula.

2. Geospiza fortis.
4. Certhidea olivasea.

Evolution

- Evolution = changes in allele frequencies across generations (ultimately leading to phenotypic change, speciation, etc.)
- Natural selection = differences in survival and reproduction across phenotypes



Three conditions required for evolution by natural selection to be possible

1. Phenotypic variation in a trait

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2. Variation is linked to fitness

Three conditions required for evolution by natural selection to be possible

1. Phenotypic variation in a trait
2. Variation is linked to fitness
3. Trait is heritable

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Is behavior "special" (e.g., relative to other phenotypes)?

Why study behavioral evolution?

- Behavior presents challenges to conventional evolutionary thinking

"... but will confine myself to one special difficulty, which at first appeared to me insuperable, and actually fatal to my whole theory. I allude to the neuters or sterile females in insect-communities”

– Charles Darwin, On the Origin of Species

Why study behavioral evolution?

- Behavior presents challenges to conventional evolutionary thinking
- Behavior can be both a result and a driver of evolutionary change

Why study behavioral evolution?

- Behavior presents challenges to conventional evolutionary thinking
- Behavior can be both a result and a driver of evolutionary change
- Animal behavior can have massive impacts on ecosystem services, agriculture, etc.

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How can we study behavioral evolution?

- Identify variation in behavior

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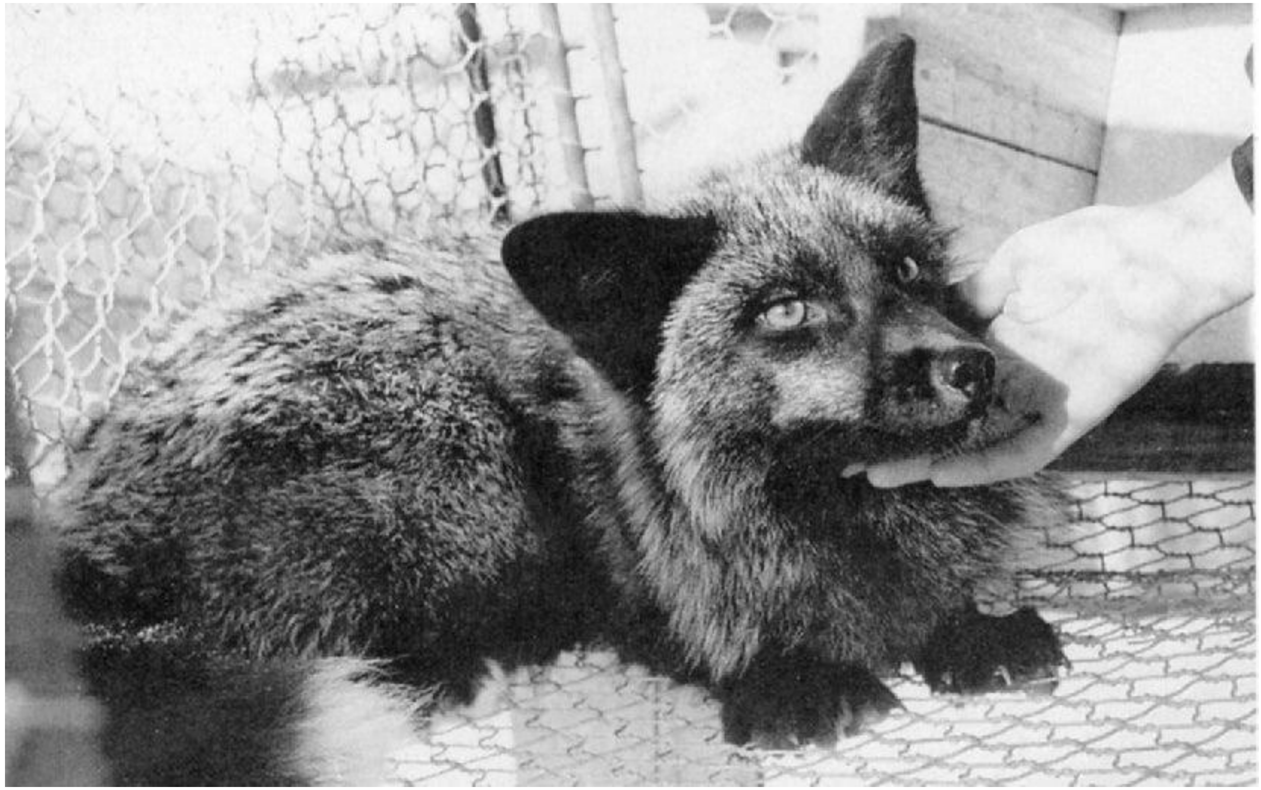
- **Hypothesize connection with ecological function, environmental response, etc. (fitness) and design study to compare behavior across conditions**

How can we study behavioral evolution?

- Identify variation in behavior
- Hypothesize connection with ecological function, environmental response, etc. (fitness) and design study to compare behavior across conditions
- **Measure heritability of behavior**

Measuring heritability of behavior

- **Selection experiments**
  - Breeding individuals with extreme traits



Measuring heritability of behavior

- Selection experiments
- **Common garden experiments**
  - Genetically distinct individuals reared in identical environment
  - Any behavioral differences are likely due to genetic variation

Genetic mechanisms of behavior

- Behavior is rarely driven by one gene or locus
- Often multiple genes
- Behavior often sensitive to changes in gene expression
  - Will cover in more detail in Week 13

The “phenotypic gambit”

- Assumption that we can use the fitness of phenotypes to predict behavioral evolution without worrying about genetic mechanisms
  - More focus on ecological factors that are shaping behavior

Using fossils to study behavioral evolution

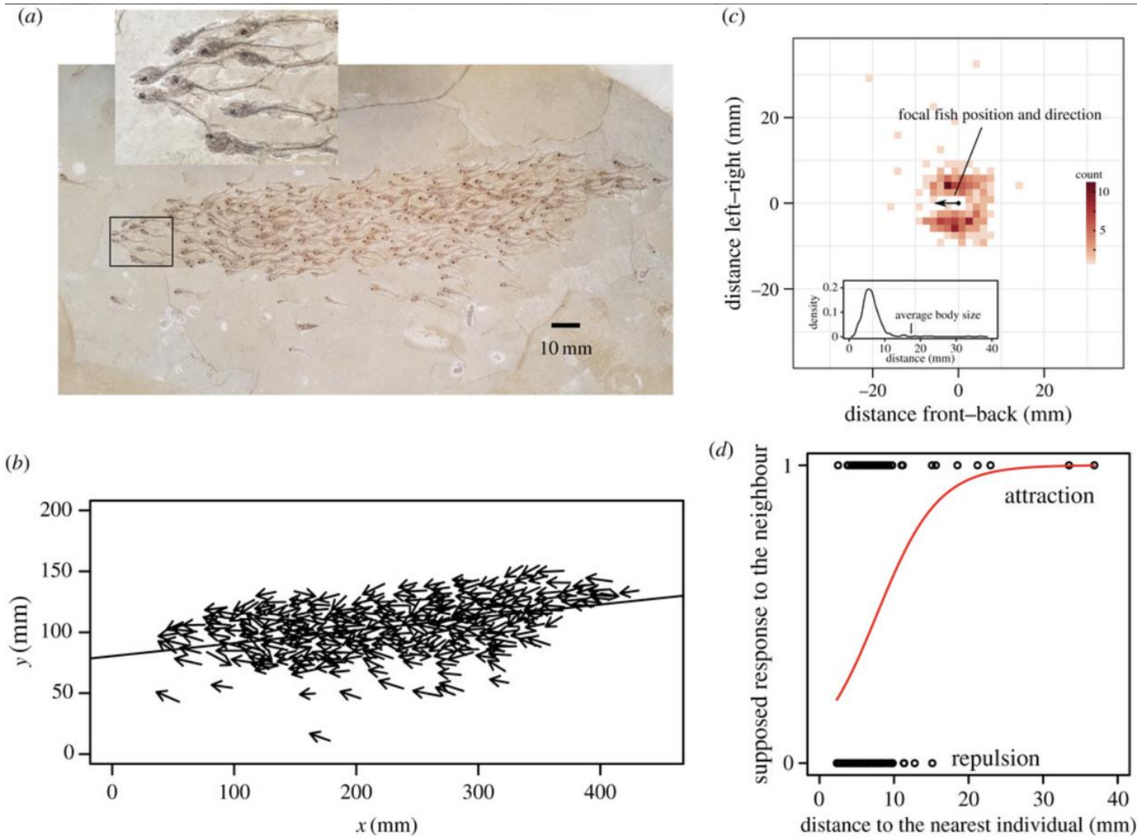
Research articles

Inferring collective behaviour from a fossilized fish shoal

Nobuaki Mizumoto, Shinya Miyata and Stephen C. Pratt

Published: **29 May 2019** https://doi.org/10.1098/rspb.2019.0891

Review history



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Honeybees

- Explored relationship between gene expression and worker task (in-nest vs foraging)

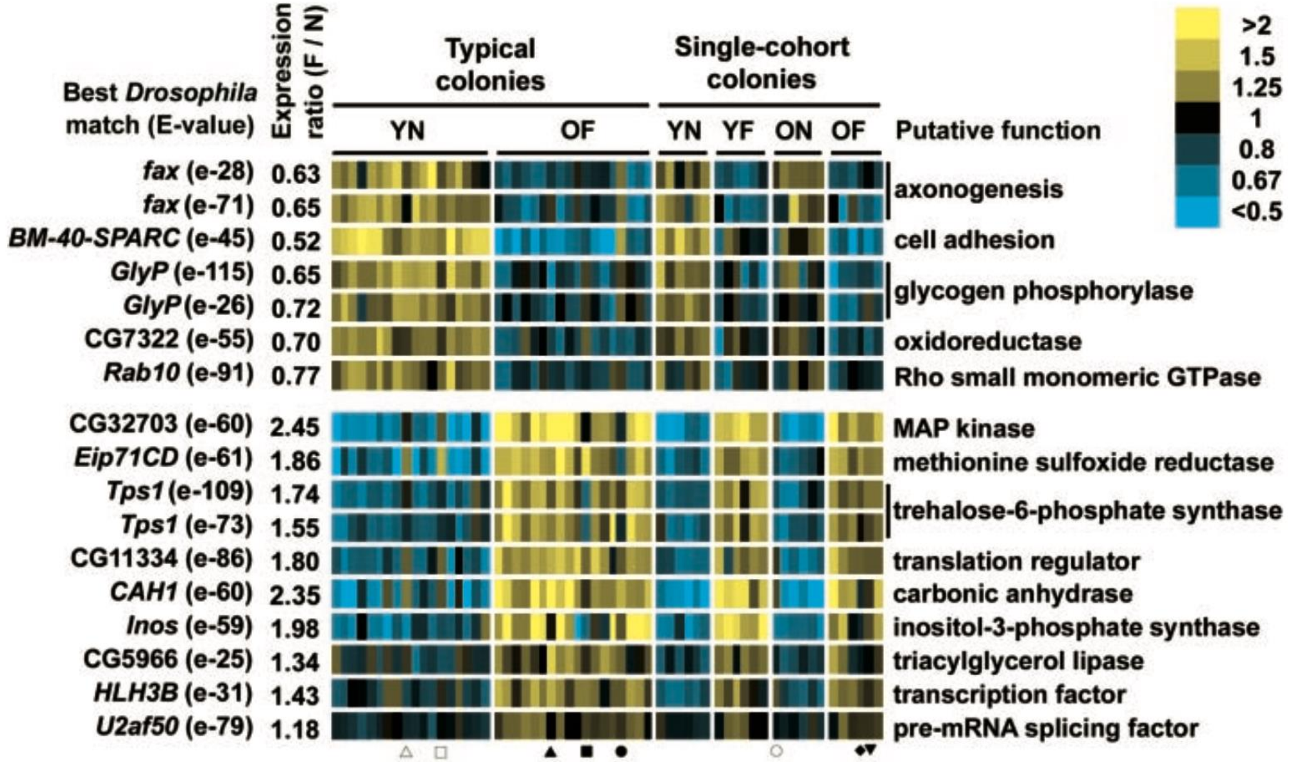
Gene Expression Profiles in the Brain Predict Behavior in Individual Honey Bees

Charles W. Whitfield,[1,2] Anne-Marie Cziko,[1] Gene E. Robinson[1,2*]

Whitfield et al 2003

Honeybees

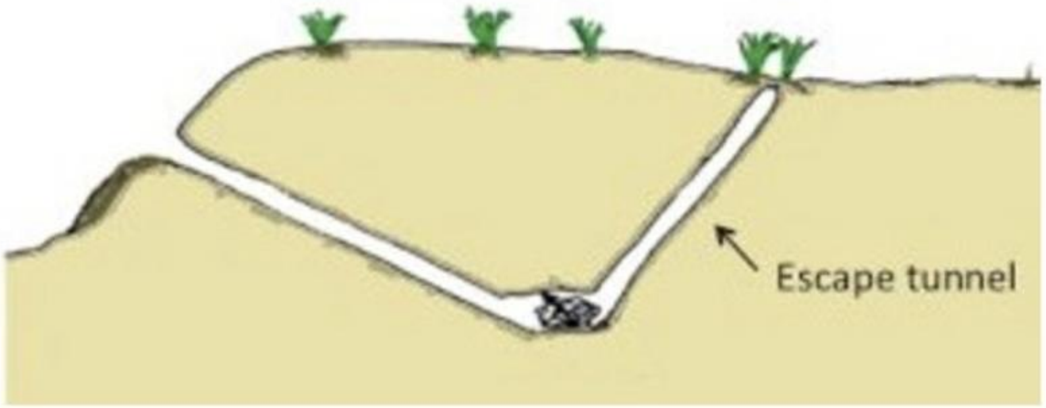
- Transition from hive-related tasks to foraging linked to expression of a suite of genes
- Gene expression predicted behavior of worker in 57/60 samples

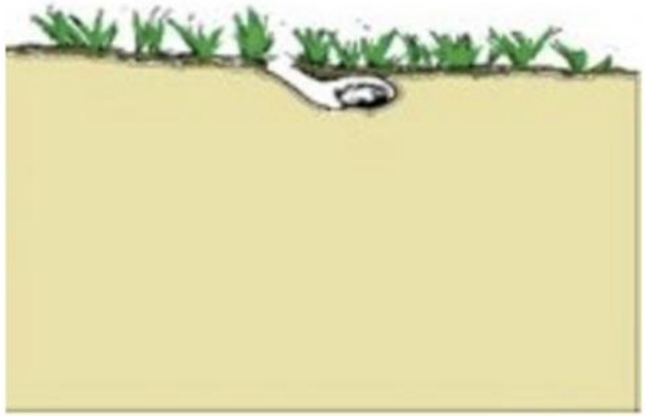


Whitfield et al 2003

Deer mice

- Deer mice (*Peromyscus*)
  - *P. maniculatus* = short, simple burrows
  - *P. polionotus* = long, elaborate burrows





Harringmeyer et al 2023

Deer mice

- Quantitative trait locus (QTL) study found forelimb digging and hindlimb kicking were associated with a 12 mB locus
- Could behavior be associated with limb morphology?

Harringmeyer et al 2023

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Leading class discussions

- As discussion leader:
  - Read paper in detail and answer instructor’s questions
  - Generate 4-5 discussion questions
    - These should be open-ended questions that promote critical analysis and discussion of the paper, not just asking about specific details
  - Lead discussion of paper with small group during class
  - Upload your questions to Canvas before class* under the “Leading in-class discussion” assignment
    - *Except for today’s discussion leaders

Leading class discussions

2025 - Fall

- Home
- Announcements
- Assignments
- Grades
- Discussions
- Badges
- Office 365
- Course Gallery - Kaltura at Illinois
- Day1Access Course Materials

▾ **Undated Assignments**

Seminar review

Leading in-class discussion #1
-/50 pts

Leading in-class discussion #2
-/50 pts

Leading in-class discussion #3
-/50 pts

Leading in-class discussion #4
-/50 pts

Class participation
-/200 pts

Leading class discussions

- All readings are uploaded to Canvas
- Questions associated with readings will be posted at least a week before that class

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